

Technical Report Describing Test Scenarios - Milestone 1

Project: Rolling Loans: Continuous Borrowing on Danogo

Milestone 1: Smart Contract Development

Evidence scope: detailed test results, technical report describing test scenarios, testnet deployment transaction evidence, and Preview/Preprod transaction IDs demonstrating functionality.

1. Purpose

This technical report describes the test scenarios used to demonstrate the Milestone 1 smart contract behavior. The focus is the core rolling-loan primitive: moving an active loan state into a new loan state through a single atomic refinancing transaction.

2. Scenario Descriptions

Scenario 1 - Rolling-loan state transition design

The rolling-loan design is modeled as a state transition from an existing active loan into a new active loan. The refinancing transaction creates a new loan, settles the existing loan, and transfers collateral into the new position in one transaction.

Scenario 2 - Source loan state before migration

The source loan state is represented by input UTxO `fc25ba0804c0fc8cb2552b97695b6ea6ec64f21551bebccd49e71b5c2abb2249#1`. This UTxO is locked at the source loan script address and contains a datum hash and parsed datum fields for loan identifier, borrowed token, loan amount, and initial interest index.

Scenario 3 - Target loan state after migration

The target loan state is represented by an output UTxO at the target loan script address. The output contains a new datum hash and parsed datum fields showing `fUSDM` as the borrowed token, an updated loan amount, and an updated initial interest index.

Scenario 4 - Atomic create + repay + collateral transfer

The execution transaction performs Migrate Loan and Create Loan in the same transaction. This means the old loan is not manually closed first; the old position is settled atomically using the newly originated loan, while collateral is redirected into the new position.

Scenario 5 - Interest and baseline handling

The target loan datum starts from a new initial interest index. This aligns with the business rule that the new loan starts with the current pool interest index, segmenting previous interest from the new term and avoiding double-counting.

Scenario 6 - Fixed/floating relevance

The tested path uses Float Loan SC as the source and Leverage Loan SC as the target. The purpose is to validate the common contract-level refinancing primitive used by supported pool-to-pool transition paths. Fixed and floating behavior is parameterized through pool configuration and interest parameters.

3. Transaction-Level Walkthrough

Step	Evidence	Interpretation
1. Source loan input	<code>fc25ba0804c0fc8cb2552b97695b6ea6ec64f21551bebccd49e71b5c2abb2249#1</code>	Existing loan UTxO is consumed by the rolling-loan transaction.
2. Target loan output	<code>addr_test1zpk4s8x6yjl8wspfqheqdgtr89dzemyrpqyc7n5my9v8027kk6djsz5vkduazxj0d7avth6k52cqy0ld77y64tpnllyesxp2fur</code>	New loan UTxO is created at the target script address.
3. Token continuity	<code>fUSDM</code> before and after	Borrowed asset remains consistent across the transition.
4. Loan amount update	<code>50,000,000 -> 50,011,310</code>	The target state carries the loan forward with updated amount.
5. Interest baseline update	<code>1,010,292,917,893 -> 1,061,924,584,989</code>	The target state starts from a refreshed interest baseline.

4. Relationship Between Aiken Unit Tests and Preprod Evidence

The Aiken file `tests/rolling_loan_test.ak` provides fixture-based unit tests for the same rolling-loan/refinancing mechanism described in this report.

The unit tests validate the contract constraints using controlled fixture values, including:

- successful full refinance flow

- interest-index update after elapsed time
- interest baseline reset on the new loan
- collateral transfer / continuity
- pool total borrow update
- undistributed fee update
- total supply update
- interest time update
- borrow APY recalculation
- utilization cap enforcement
- loan amount and token consistency
- invalid collateral rejection
- pool reserve constraint validation

The Preprod transaction evidence demonstrates the same mechanism with actual on-chain values. Therefore, the unit test file and the transaction evidence serve complementary purposes: the .ak file shows test-level constraint coverage, while the Preprod transaction shows the deployed behavior accepted on-chain.

5. Conclusion

The tested flow demonstrates an atomic rolling-loan/refinancing state transition. It avoids the user flow of manually closing the old loan, withdrawing collateral, and opening a new loan in separate steps. Instead, new-loan creation, old-loan settlement, and collateral transfer occur in one transaction.